
MentorLink Database: Phase II



By: Arianna Kelsey, Mathieu Poulin, Hanan Ali, Jordie Beresin

Data Scraping

[Apply](#) [Visit](#) [Get Info](#) [Give](#) [MyUSM](#)

 UNIVERSITY OF
SOUTHERN MAINE

[Academics](#) [Admissions & Aid](#) [Student Life](#)  

University Faculty



**Abou El-Makarim
Aboueissa, PhD**
HE | HIM | HIS
Professor, Department of
Mathematics and Statistics
207-228-8389
aboue.aboueissa@maine.edu



Titus Abbott
HE | HIM | HIS
Artist Faculty in Music
Theory & Composition
titus.abbott@maine.edu



Anne Ackerman
SHE | HER | HERS
Part-Time Lecturer of Art
Education
anne.ackerman@maine.edu



**Valerie Ackley,
MA**
SHE | HER | HERS



Faith Adiele
SHE | HER | HERS
Stonecoast MFA Faculty



Elvice Adji, MS
HE | HIM | HIS
Lecturer III, Part Time

```
[
{
  "source_url": "https://usm.maine.edu/directories/people/abou-el-makarim-aboueissa/",
  "full_name": "Abou El-Makarim Aboueissa, PhD",
  "email": "abouel.aboueissa@maine.edu",
  "phone": "207-228-8389",
  "raw_bio_text": "Abou El-Makarim Aboueissa, PhD HE | HIM | HIS Professor, Department of Mathematics and Statistics 207-228-8389 abouel.aboueissa@maine.edu Science Building, room C581, C-wing, Portland campus Education PhD, Statistics, Western Michigan University MS, Applied Statistics, Michigan State University MS, Mathematical Statistics, Alexandria University BS, Statistics Department of Mathematics and Statistics College of Technology, Science, and Health Professor Aboueissa's teaching experience includes teaching both undergraduate and graduate courses in statistics, biostatistics, data science, mathematics, and statistical evaluation of the clinical data from multi-center clinical trials. Professor Aboueissa has a good range of teaching experience both in person and online gained over 35 years. Expertise Professor Aboueissa's training in theoretical statistics coupled with over 20 years of collaborations in the areas of public health, medicine, nursing, applied medical, pharmacology, and communica"
},
{
  "source_url": "https://usm.maine.edu/directories/people/titus-abbott/",
  "full_name": "Titus Abbott",
  "email": "titus.abbott@maine.edu",
  "phone": null,
  "raw_bio_text": "Titus Abbott HE | HIM | HIS Artist Faculty in Music Theory & Composition titus.abbott@maine.edu Education MM, Composition, University of Southern Maine BM, Classical Composition, University of Montreal Osher School of Music College of Arts, Humanities, and Social Sciences Titus Abbott, saxophonist and composer in both jazz and classical idioms has been performing nationally and internationally for over 25 years. As an improviser and composer, he has worked and recorded with a wide range of collaborators from his native Canada to Europe and the US. These include Graham Collier, Eddie Pévost, The Grand Union Orchestra, George Koller, Al Gardner, Frank Carlberg, Pascal Niggenkemper, Michael Sarin, Marilyn Lerner, Curtis Clark, Nate McBride, Luther Gray, Sean Conly and many others performing at venues, concert halls and jazz festivals worldwide. New music composition and film music composition are also part of the fabric of Titus' career which include composer credits on films by Louis Pas"
},
{
  "source_url": "https://usm.maine.edu/directories/people/anne-ackerman/",
  "full_name": "Anne Ackerman",
  "email": "anne.ackerman@maine.edu",
  "phone": null,
  "raw_bio_text": "Anne Ackerman SHE | HER | HERS Part-Time Lecturer of Art Education anne.ackerman@maine.edu Download Curriculum Vitae Robie-Andrews Hall, room 109, Gorham campus Education MEd in Teacher Leadership, University of Southern Maine BFA in Art Education, University of Southern Maine BA in Art, University of Southern Maine Department of Art College of Arts, Humanities, and Social Sciences anne.ackerman@maine.edu Download Curriculum Vitae Robie-Andrews Hall, room 109, Gorham campus Education MEd in Teacher Leadership, University of Southern Maine BFA in Art Education, University of Southern Maine BA in Art, University of Southern Maine"
},
{
  "source_url": "https://usm.maine.edu/directories/people/valerie-ackley/",
  "full_name": "Valerie Ackley, MA",
  "email": "valerie.ackley@maine.edu",
  "phone": "207-240-1599",
  "raw_bio_text": "Valerie Ackley, MA SHE | HER | HERS Lecturer II, Part Time Faculty, Department of Mathematics and Statistics 207-240-1599 (text only) valerie.ackley@maine.edu Part-time Instructors' Office, Lewiston campus Education MA, Curriculum/Instruction with PK-14 Mathematics Focus BA, Mathematics/Secondary Education Department of Mathematics and Statistics College of Science, Technology, and Health 207-240-1599 (text only) valerie.ackley@maine.edu Part-time Instructors' Office, Lewiston campus Education MA, Curriculum/Instruction with PK-14 Mathematics Focus BA, Mathematics/Secondary Education"
},
{
  "source_url": "https://usm.maine.edu/directories/people/faith-adiele/",
  "full_name": "Faith Adiele",
  "email": "faithe.adiele@maine.edu",
  "phone": "207-780-4423",
  "raw_bio_text": "Faith Adiele SHE | HER | HERS Stonecoast MFA Faculty Stonecoast MFA Office: 207-780-4423 faithe.adiele@maine.edu Education MFA Fiction, Iowa Writers Workshop MFA Creative Nonfiction, Iowa Nonfiction Program MA Creative Writing, Lesley University BA Southeast Asian Studies, Harvard University Faith Adiele is the author of Meeting Faith , a dual narrative about becoming Thailand's first Black Buddhist Nun that won the PEN Open Book Award for Memoir, routinely makes Best Of lists, and is widely taught in American universities. She has also published four experimental chapbooks about her Nigerian-Nordic-American heritage – the audio/e-book, The Nigerian-Nordic Girl's Guide to Lady Problems , a collaboration with a Kenyan graphic artist, and recently, two hybrid, speculative memoirs: Her Voice: Hänen Äänensä: A Hybrid Memoir and Voice/Over: A Memoir Breakout in 7 Movies. Her media credits include My Journen Home (a PBS documentary about finding her father and siblings in Nigeria). two e"
```

	A	B	C	D	E	G
1	FirstName	LastName	Title	Department	Email	PhoneNumber
2	Abou	El-Makarim Aboueissa, PhD	Professor	Department of Mathematics and Statistics	abouel.aboueissa@maine.edu	207-228-8389
3	Titus	Abbott			titus.abbott@maine.edu	
4	Anne	Ackerman	Lecturer		anne.ackerman@maine.edu	
5	Valerie	Ackley, MA	Lecturer	Department of Mathematics and Statistics	valerie.ackley@maine.edu	207-240-1599
6	Faith	Adiele			faithe.adiele@maine.edu	207-780-4423
7	Elvice	Adji, MS	Lecturer	Department of Mathematics and Statistics	elvice.adji@maine.edu	207-780-4608
8	Katherine	Ahrens, PhD	Professor		katherine.ahrens@maine.edu	207-780-4847
9	Janna	Ahrndt	Assistant Professor	Art	janna.ahrndt@maine.edu	207-780-5462
10	Heather	Alden MEd			heather.alden@maine.edu	207-780-5371
11	Dami	Alegbeleye, PhD	Assistant Professor	College of Arts	ibukun.alegbeleye@maine.edu	207-780-5047
12	JJ	Amaworo Wilson			jeremy.j.wilson@maine.edu	207-780-4423
13	Chris	Ambrose			christopher.ambrose@maine.edu	(207)780-4716
14	Eklou	Amendah	Assistant Professor	Marketing	eklou.amendah@maine.edu	207-780-4595
15	Amy	Amoroso, MFA			amy.amoroso@maine.edu	207-780-4321
16	Jeanette	Andonian, PhD, LCSW	Associate Professor		jeanette.andonian@maine.edu	207-780-4115
17	Andrew	Hughen	Lecturer		andrew.hughen@maine.edu	207-780-4680
18	Anastasia	Antonacos	Lecturer		anastasia.antonacos@maine.edu	207-780-5778
19	Jono	Anzalone EdD	Lecturer		jono.anzalone@maine.edu	402-871-4070
20	Carolyn	Arcand	Assistant Professor	School of Public Service	carolyn.arcand@maine.edu	207-780-4615
21	Joseph	Arel	Lecturer		joseph.arel@maine.edu	207-780-4258
22	Richard	Arend	Professor		richard.arend@maine.edu	207-780-4597
23	Sara	Armas	Lecturer		sara.armas@maine.edu	
24	Pearl	Armstrong, MS	Lecturer	Department of Mathematics and Statistics	pearl.armstrong@maine.edu	207-780-4608
25	Joseph	Arsenault, MA	Lecturer	Department of Mathematics and Statistics	joseph.arsenaultjr@maine.edu	207-780-4608
26	Vahideh	Babalou			vahideh.babalou@maine.edu	
27	Roxanne	Baker			roxanne.baker1@maine.edu	
28	Matthew	Bampton	Chair	Anthropology	bampton@maine.edu	207-780-5184
29	Tabitha	Barnard	Lecturer		tabitha.barnard@maine.edu	
30	Adele	Baruch, PhD	Associate Professor		adele.baruchrunyon@maine.edu	207-780-5317
31	Rick	Bass				
32	Mary	Bastoni			mary.bastoni@maine.edu	207-780-5265
33	Kate	Beever, MA, MT-BC			kate.beever@maine.edu	
34	Alex	Beggs			alex.beggs@maine.edu	207-780-4260
35	Piers	Beirne, PhD	Professor		beirne@maine.edu	612-019-0946
36	Barbara	Belik	Lecturer		barbara.belik@maine.edu	

Schema - Mathieu

```
1 USE MentorLink;
2
3 SET FOREIGN_KEY_CHECKS = 0;
4
5 DROP TABLE IF EXISTS Session;
6
7 SET FOREIGN_KEY_CHECKS = 1;
8
9 CREATE TABLE Session (
10     SessionID BINARY(16) DEFAULT (UUID_TO_BIN(UUID())),
11     -- Added Mentorship ID
12     MentorshipID BINARY(16) NOT NULL,
13     Timestamp TIMESTAMP NOT NULL,
14     InstructionType ENUM('Virtual', 'In-Person') NOT NULL,
15     Location TINYTEXT NOT NULL,
16     -- Fixed: Added session status column. Supports CancelSession and RescheduleSession
17     -- However, will need to update ER Diagram and Data Dictionary
18     Status ENUM('Scheduled', 'Cancelled', 'Completed') NOT NULL DEFAULT 'Scheduled',
19
20     -- Added constraint style to keep with consistency and readability
21     CONSTRAINT pk_session
22     PRIMARY KEY (SessionID),
23
24     CONSTRAINT fk_session_mentorship
25     FOREIGN KEY (MentorshipID)
26     REFERENCES Mentorship(MentorshipID)
27     ON DELETE CASCADE
28     ON UPDATE CASCADE
29 );
30
31 -- Added: To help with procedures
32 CREATE INDEX idx_session_mentorship
33 ON Session(MentorshipID);
34
35 CREATE INDEX idx_session_timestamp
36 ON Session(Timestamp);
37
38 CREATE INDEX idx_session_mentorship_timestamp
39 ON Session(MentorshipID, Timestamp);
40
41 CREATE INDEX idx_session_status
42 ON Session(Status);
43
```


User Functions - Mathieu

```
1 DELIMITER //
2
3 CREATE PROCEDURE CancelSession(
4     IN p_SessionID BINARY(16)
5 )
6 BEGIN
7     DECLARE v_SessionStatus VARCHAR(20);
8     DECLARE v_RowCount INT;
9
10    -- Check if session ID is NULL
11    IF p_SessionID IS NULL THEN
12        SIGNAL SQLSTATE '45000'
13        SET MESSAGE_TEXT = 'Error: SessionID cannot be NULL';
14    END IF;
15
16    -- Check if session exists and get its current status
17    SELECT Status INTO v_SessionStatus
18    FROM Session
19    WHERE SessionID = p_SessionID;
20
21    IF v_SessionStatus IS NULL THEN
22        SIGNAL SQLSTATE '45000'
23        SET MESSAGE_TEXT = 'Error: Session not found';
24    END IF;
25
26    -- Check if session is already cancelled or completed
27    IF v_SessionStatus = 'Cancelled' THEN
28        SIGNAL SQLSTATE '45000'
29        SET MESSAGE_TEXT = 'Error: Session is already cancelled';
30    END IF;
31
32    IF v_SessionStatus = 'Completed' THEN
33        SIGNAL SQLSTATE '45000'
34        SET MESSAGE_TEXT = 'Error: Cannot cancel a completed session';
35    END IF;
36
37    -- Update session status to Cancelled
38    UPDATE Session
39    SET Status = 'Cancelled'
40    WHERE SessionID = p_SessionID;
41
42    SELECT 'Session cancelled successfully' AS Message;
43 END //
44
45 DELIMITER ;
```

```
1 DELIMITER //
2
3 CREATE FUNCTION CountAchievedGoals(
4     IN p_MentorshipID BINARY(16)
5 )
6 RETURNS INT
7 DETERMINISTIC
8 READS SQL DATA
9 BEGIN
10    DECLARE v_Count INT DEFAULT 0;
11    DECLARE v_Exists INT;
12
13    -- Check if mentorship ID is NULL
14    IF p_MentorshipID IS NULL THEN
15        RETURN 0;
16    END IF;
17
18    -- Check if mentorship exists
19    SELECT COUNT(*) INTO v_Exists
20    FROM Mentorship
21    WHERE MentorshipID = p_MentorshipID;
22
23    IF v_Exists = 0 THEN
24        RETURN 0;
25    END IF;
26
27    -- Count achieved goals
28    SELECT COUNT(*) INTO v_Count
29    FROM Goal
30    WHERE MentorshipID = p_MentorshipID
31    AND Status = 'Achieved';
32
33    RETURN v_Count;
34 END //
35
36 DELIMITER ;
```

```
1 DELIMITER //
2
3 CREATE PROCEDURE EndMentorship(
4     IN p_MentorshipID BINARY(16)
5 )
6 BEGIN
7     DECLARE v_MentorshipStatus VARCHAR(20);
8
9    -- Check if mentorship ID is NULL
10    IF p_MentorshipID IS NULL THEN
11        SIGNAL SQLSTATE '45000'
12        SET MESSAGE_TEXT = 'Error: MentorshipID cannot be NULL';
13    END IF;
14
15    -- Check if mentorship exists and get its current status
16    SELECT Status INTO v_MentorshipStatus
17    FROM Mentorship
18    WHERE MentorshipID = p_MentorshipID;
19
20    IF v_MentorshipStatus IS NULL THEN
21        SIGNAL SQLSTATE '45000'
22        SET MESSAGE_TEXT = 'Error: Mentorship not found';
23    END IF;
24
25    -- Check if mentorship is already ended
26    IF v_MentorshipStatus = 'Ended' THEN
27        SIGNAL SQLSTATE '45000'
28        SET MESSAGE_TEXT = 'Error: Mentorship is already ended';
29    END IF;
30
31    -- Update mentorship status to Ended
32    UPDATE Mentorship
33    SET Status = 'Ended'
34    WHERE MentorshipID = p_MentorshipID;
35
36    SELECT 'Mentorship ended successfully' AS Message;
37 END //
38
39 DELIMITER ;
```

```
1 DELIMITER //
2
3 CREATE PROCEDURE GetMentorshipSummary(
4     IN p_MentorshipID BINARY(16)
5 )
6 BEGIN
7     DECLARE v_Exists INT;
8
9    -- Check if mentorship ID is NULL
10    IF p_MentorshipID IS NULL THEN
11        SIGNAL SQLSTATE '45000'
12        SET MESSAGE_TEXT = 'Error: MentorshipID cannot be NULL';
13    END IF;
14
15    -- Check if mentorship exists
16    SELECT COUNT(*) INTO v_Exists
17    FROM Mentorship
18    WHERE MentorshipID = p_MentorshipID;
19
20    IF v_Exists = 0 THEN
21        SIGNAL SQLSTATE '45000'
22        SET MESSAGE_TEXT = 'Error: Mentorship not found';
23    END IF;
24
25    -- Return comprehensive mentorship summary
26    SELECT
27        p_MentorshipID,
28        m.SubjectName,
29        m.Status AS MentorshipStatus,
30        mtor_u.UserID AS MentorID,
31        mtor_u.FirstName AS MentorFirstName,
32        mtor_u.LastName AS MentorLastName,
33        mtor_g.GoalID AS MentorGoalID,
34        mtor_g.UserID AS MentorID,
35        mtor_g.FirstName AS MentorFirstName,
36        mtor_g.LastName AS MentorLastName,
37        mtor_g.GoalID AS MentorGoalID,
38        mtor_g.GoalStatus AS MentorGoalStatus,
39        COALESCE(gol_counts.TotalGoals, 0) AS TotalGoals,
40        COALESCE(gol_counts.AchievedGoals, 0) AS AchievedGoals,
41        COALESCE(gol_counts.CompletedSessions, 0) AS CompletedSessions
42    FROM Mentorship m
43    INNER JOIN Subject s ON m.SubjectID = s.SubjectID
44    LEFT JOIN MentorshipMentor m_mtor ON m.MentorshipID = m_mtor.MentorshipID
45    AND m_mtor.RoleName = 'Mentor'
46    LEFT JOIN User mtor_u ON m_mtor.UserID = mtor_u.UserID
47    LEFT JOIN MentorshipMentor m_gtor ON m.MentorshipID = m_gtor.MentorshipID
48    AND m_gtor.RoleName = 'Mentee'
49    LEFT JOIN User mtor_g ON m_gtor.UserID = mtor_g.UserID
50    LEFT JOIN Goal mtor_g ON m_gtor.MentorshipID = mtor_g.MentorshipID
51    SELECT
52        m.MentorshipID,
53        COUNT(*) AS TotalGoals,
54        SUM(CASE WHEN Status = 'Achieved' THEN 1 ELSE 0 END) AS AchievedGoals
55    FROM Goal
56    GROUP BY MentorshipID
57    gol_counts ON m.MentorshipID = gol_counts.MentorshipID
58    LEFT JOIN
59    SELECT
60        MentorshipID,
61        COUNT(*) AS CompletedSessions
62    FROM Session
63    WHERE Status = 'Completed'
64    GROUP BY MentorshipID
65    session_counts ON m.MentorshipID = session_counts.MentorshipID
66    WHERE m.MentorshipID = p_MentorshipID;
67 END //
68
69 DELIMITER ;
```



SOUTHERN MAINE

COMMUNITY COLLEGE

[Apply](#)
[Get Info](#)
[Give](#)
[MyUMC](#)

[Academics](#)
[Admissions & Aid](#)
[Student Life](#)




Academic Catalog

Catalog Search

[Catalog Search](#)

[Whole Word/Phrase](#)

[Advanced Search](#)

[Catalog Home](#)

[President's Welcome](#)

[University Overview](#)

[Academic Policies](#)

[Admissions](#)

[Academic Support Services](#)

[Campus Life](#)

[Core Curriculum](#)

[Disclaimers](#)

[Financial Aid](#)

[Financial Information](#)

[Registration & Scheduling Services](#)

[Programs by College/Department](#)

[Programs \(A-Z\)](#)

2025-2026 Undergraduate Catalog




Course Descriptions

List of Course Prefixes

Course Description by Department/Program

Course Filter

Filter this list of courses using course prefix, course code, keywords or any combination.

Prefix:	Code or Number:	Type:	Keyword or Phrase:
All prefixes		All types	Filter

☐ Find whole word or phrase only.

Accounting

- ACC 105 - Show Me the Money
- ACC 110 - Financial Accounting Information for Decision Making
- ACC 211 - Managerial Accounting Information for Decision Making
- ACC 295 - Internship in Accounting
- ACC 301 - Financial Reporting I
- ACC 302 - Financial Reporting II
- ACC 315 - Estate Planning
- ACC 329 - Accounting Information Systems
- ACC 395 - Internship I
- ACC 396 - Internship II
- ACC 405 - Cost Management Systems
- ACC 410 - Auditing and Assurance
- ACC 413 - Concepts and Strategies of Taxation
- ACC 416 - Governmental and Nonprofit Accounting
- ACC 418 - Principles of Fraud Examination
- ACC 490 - Independent Study in Accounting
- ACC 499 - Special Topics in Accounting

Accounting & Finance

```

2      {
3      "CourseName": "ACC 185 -- Show Me the Money"
4      },
5      {
6      "CourseName": "ACC 110 -- Financial Accounting Information for Decision Making"
7      },
8      {
9      "CourseName": "ACC 211 -- Managerial Accounting Information for Decision Making"
10     },
11     {
12     "CourseName": "ACC 295 -- Internship in Accounting"
13     },
14     {
15     "CourseName": "ACC 381 -- Financial Reporting I"
16     },
17     {
18     "CourseName": "ACC 382 -- Financial Reporting II"
19     },
20     {
21     "CourseName": "ACC 315 -- Estate Planning"
22     },
23     {
24     "CourseName": "ACC 329 -- Accounting Information Systems"
25     },
26     {
27     "CourseName": "ACC 395 -- Internship I"
28     },
29     {
30     "CourseName": "ACC 396 -- Internship II"
31     },
32     {
33     "CourseName": "ACC 485 -- Cost Management Systems"
34     },
35     {
36     "CourseName": "ACC 418 -- Auditing and Assurance"
37     },
38     {
39     "CourseName": "ACC 413 -- Concepts and Strategies of Taxation"
40     },
41     {
42     "CourseName": "ACC 416 -- Governmental and Nonprofit Accounting"
43     },
44     {
45     "CourseName": "ACC 418 -- Principles of Fraud Examination"
46     },
47     {
48     "CourseName": "ACC 498 -- Independent Study in Accounting"
49     },
50     {
51     "CourseName": "ACC 499 -- Special Topics in Accounting"
52     }
53     ]
54 }

```

A	B	C	D
1	CourseName	Department	
2	ACC 105 - Show Me the Money	Accounting	
3	ACC 110 - Financial Accounting Information for Decision Making	Accounting	
4	ACC 211 - Managerial Accounting Information for Decision Making	Accounting	
5	ACC 295 - Internship in Accounting	Accounting	
6	ACC 301 - Financial Reporting I	Accounting	
7	ACC 302 - Financial Reporting II	Accounting	
8	ACC 315 - Estate Planning	Accounting	
9	ACC 329 - Accounting Information Systems	Accounting	
10	ACC 395 - Internship I	Accounting	
11	ACC 396 - Internship II	Accounting	
12	ACC 405 - Cost Management Systems	Accounting	
13	ACC 410 - Auditing and Assurance	Accounting	
14	ACC 413 - Concepts and Strategies of Taxation	Accounting	
15	ACC 416 - Governmental and Nonprofit Accounting	Accounting	
16	ACC 418 - Principles of Fraud Examination	Accounting	
17	ACC 490 - Independent Study in Accounting	Accounting	
18	ACC 499 - Special Topics in Accounting	Accounting	
19	FIN 201 - Personal Finance	Accounting & Finance	
20	FIN 295 - Internship in Finance	Accounting & Finance	
21	FIN 299 - Special Topics in Finance	Accounting & Finance	
22	FIN 315 - Estate Planning	Accounting & Finance	
23	FIN 320 - Basic Financial Management	Accounting & Finance	
24	FIN 321 - Personal Financial Planning	Accounting & Finance	
25	FIN 323 - Financial Engineering	Accounting & Finance	
26	FIN 326 - Financial Modeling	Accounting & Finance	
27	FIN 327 - Investment Management	Accounting & Finance	
28	FIN 328 - Financial Markets and Institutions	Accounting & Finance	
29	FIN 330 - International Financial Management	Accounting & Finance	
30	FIN 395 - Internship I	Accounting & Finance	
31	FIN 396 - Internship II	Accounting & Finance	
32	FIN 399 - Special Topics in Finance	Accounting & Finance	
33	FIN 490 - Independent Study in Finance	Accounting & Finance	
34	AFL 001 - Leadership Laboratory	Air Force ROTC	
35	AFL 115 - Heritage & Values of USAF I	Air Force ROTC	
36	AFL 116 - Heritage & Values of USAF II	Air Force ROTC	
37	AFL 241 - Team and Leadership Fundamentals I	Air Force ROTC	
38	AFL 242 - Team and Leadership Fundamentals II	Air Force ROTC	

Query Optimization* - Mathieu

*Query #4 is a pessimization of query #2

The screenshots show a query optimizer interface with a query editor, a result grid, and an action output section. The query is a SELECT statement with a WHERE clause filtering by MentorshipID = 1919 and an ORDER BY clause on Timestamp.

Query Editor (Top Left):

```
SELECT *
FROM Session
WHERE MentorshipID = 1919
ORDER BY Timestamp;
```

Result Grid (Top Right):

SessionID	MentorshipID	Timestamp	Location	Status
2026-05-05 13:30:00	In-Person	Portland, Science Building C-245	Scheduled	

Action Output (Bottom Left):

Time	Action	Response	Duration / Fetch Time
18:20:35	SELECT SessionID, MentorshipID, Timestamp...	1 row(s) returned	0.0000 sec / 0.0000...
18:21:26	EXPLAIN SELECT SessionID, Timestamp, Instru...	1 row(s) returned	0.0023 sec / 0.0000...
18:21:26	SELECT SessionID, Timestamp, InstructionID...	1 row(s) returned	0.0007 sec / 0.0000...

Query Editor (Bottom Left):

```
SELECT *
FROM Session
WHERE MentorshipID = 1919
ORDER BY Timestamp;
```

Result Grid (Bottom Right):

SessionID	MentorshipID	Timestamp	Location	Status
2026-05-05 13:30:00	In-Person	Portland, Science Building C-245	Scheduled	

Action Output (Bottom Right):

Time	Action	Response	Duration / Fetch Time
18:19:59	SELECT SessionID, MentorshipID, Timestamp...	1 row(s) returned	0.0006 sec / 0.0000...
18:19:59	EXPLAIN SELECT SessionID, MentorshipID, Tim...	1 row(s) returned	0.0008 sec / 0.0000...
18:19:59	SELECT SessionID, MentorshipID, Timestamp...	1 row(s) returned	0.0007 sec / 0.0000...

Indices - Mathieu

```
32 CREATE INDEX idx_session_mentorship
33     ON Session(MentorshipID);
34
35 CREATE INDEX idx_session_timestamp
36     ON Session(Timestamp);
37
38 CREATE INDEX idx_session_mentorship_timestamp
39     ON Session(MentorshipID, Timestamp);
40
41 CREATE INDEX idx_session_status
42     ON Session(Status);
43
```

SCHEMA - HA

```
1  USE MentorLink;
2  SET FOREIGN_KEY_CHECKS = 0;
3
4  DROP TABLE IF EXISTS User_Subject;
5  DROP TABLE IF EXISTS User;
6
7  SET FOREIGN_KEY_CHECKS = 1;
8
9  -- Create User table. PK: UserID
10 CREATE TABLE User (
11     UserID BINARY(16) NOT NULL DEFAULT (UUID_TO_BIN(UUID())),
12     FirstName VARCHAR(32) NOT NULL,
13     LastName VARCHAR(32) NOT NULL,
14     Email VARCHAR(255) NOT NULL,
15     PhoneNumber VARCHAR(20),
16     Password VARCHAR(128) NOT NULL,
17
18     CONSTRAINT pk_user PRIMARY KEY (UserID),
19     CONSTRAINT uq_user_email UNIQUE (Email)
20 );
21
22 -- Create User_Subject table.
23 CREATE TABLE User_Subject (
24     UserID BINARY(16) NOT NULL,
25     SubjectID BINARY(16) NOT NULL,
26     UserType ENUM('Sought', 'Offered') NOT NULL,
27
28     -- PK: UserID, SubjectID, UserType
29     PRIMARY KEY(UserID, SubjectID, UserType),
30
31     -- FK: UserID, SubjectID
32     CONSTRAINT fk_user_ID
33         FOREIGN KEY (UserID)
34         REFERENCES User(UserID)
35         ON DELETE CASCADE
36         ON UPDATE CASCADE,
37     CONSTRAINT fk_subject_ID
38         FOREIGN KEY (SubjectID)
39         REFERENCES Subject(SubjectID)
40         ON DELETE CASCADE
41         ON UPDATE CASCADE
42 );
43
44 -- Added: Index to help find all users for a subject and type
45 CREATE INDEX idx_usersubject_subject_type
46     ON User_Subject(SubjectID, UserType);
```

- User
- User_Subject

STORED PROCEDURES - HA

- Add Goal to Mentorship
 - Mark Goal Achieved
 - Schedule Session
 - Update Goal Description
-

DATA SCRAPING - HA

RESEARCH AREAS

LINK:

https://en.wikipedia.org/wiki/Outline_of_academic_disciplines

QUERY OPTIMIZATION - HA

-- Query 1: View the name of a subject AND its ID

-- from a specific department

-- Possible Use: Find a subject in a department

-- and view its ID to 'Add_User_Subject_Interest'

SELECT

BIN_TO_UUID(SubjectID) AS SubjectID,

SubjectName

FROM Subject

WHERE Subject.Department = 'History';

Query 1 Index

```
CREATE INDEX idx_Subject_Department
```

```
ON Subject (Department);
```

QUERY OPTIMIZATION 2 - HA

-- Query 2: Find Users that are offering subjects

-- in a specific department

-- Possible Use: Looking for individuals offering

-- mentorship for a subject

SELECT

User.FirstName,

User.LastName,

Subject.SubjectName,

User_Subject.UserType

FROM User_Subject

JOIN Subject

ON User_Subject.SubjectID = Subject.SubjectID

JOIN User

ON User.UserID = User_Subject.UserID

WHERE Subject.Department = 'History'

AND User_Subject.UserType = 'Offered';

Query 2 Index

```
CREATE INDEX idx_Subject_Department
ON Subject (SubjectID,
Department);
```

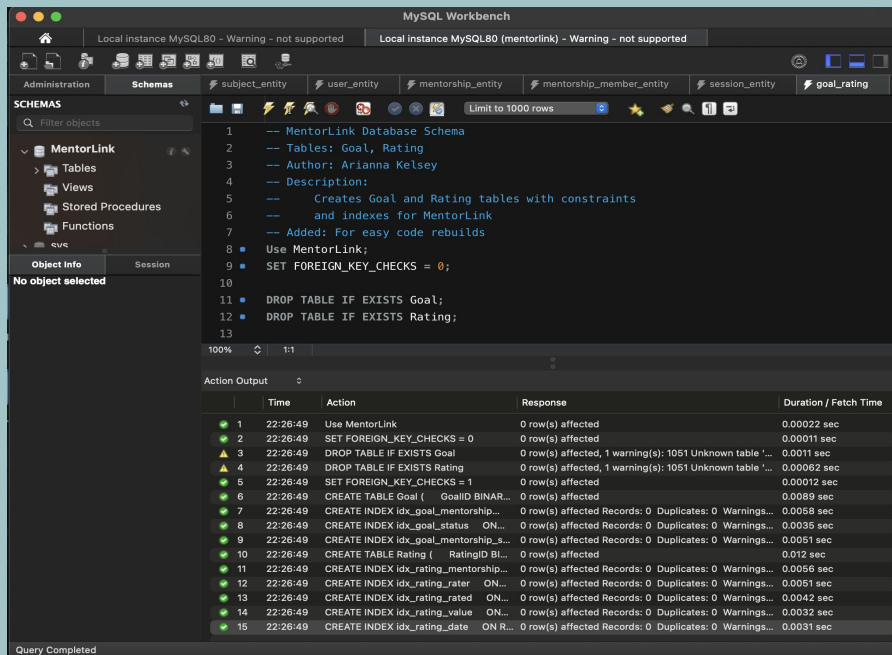
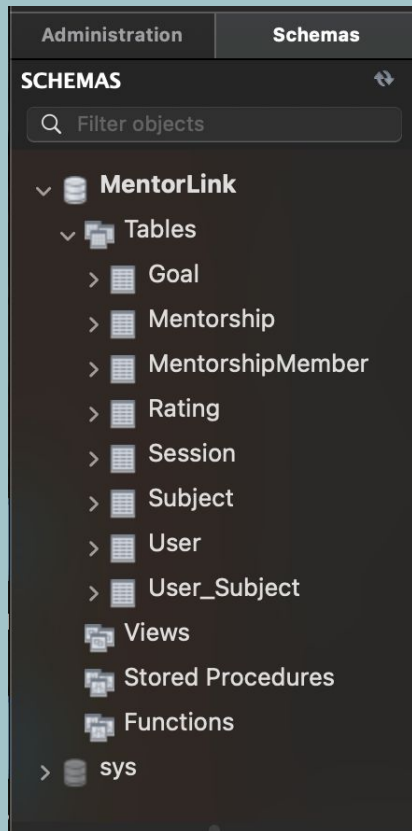
```
CREATE INDEX idx_User_Subject_Type
ON User_Subject (UserType,
SubjectID, UserID);
```

Synopsis of MentorLink Contributions

- Arianna

- Goal & Rating Tables
 - Sample Data: Goal, Rating, User-Subject
 - Procedures
 - AddUserSubject, RemoveUserSubject, SubmitRating, RescheduleSession
 - Only covering SubmitRating procedure for brevity
 - Opportunities Data Scraping/Cleaning
 - Opportunities Query Optimization and Indexing
-

Schema Scripts Running Locally



Sample Data Functionality

MySQL Workbench interface showing a query execution result for the 'Goal' table.

Query executed:

```
1 USE MentorLink;
2
3 SELECT * FROM Goal;
```

Result Grid:

GoalID	MentorshipID	Description	Status
BLOB	BLOB	Master core algebra concepts	Achieved
BLOB	BLOB	Complete five algebra practice sets	Set
BLOB	BLOB	Improve Python programming fundamentals	Achieved
BLOB	BLOB	Build a small Python project	Set
BLOB	BLOB	Edit two writing pieces for class	Achieved
BLOB	BLOB	Write a short story	Set
BLOB	BLOB	Deliver a short mock presentation	Achieved
BLOB	BLOB	Practice powerpoint skills	Set
BLOB	BLOB	Create a polished one-page resume	Achieved
BLOB	BLOB	Learn introductory SQL queries	Set
BLOB	BLOB	Review core chemistry principles	Achieved
BLOB	BLOB	Explore introductory finance concepts	Set
BLOB	BLOB	Memorize introductory biology concepts fo...	Set
BLOB	BLOB	Understand foundational psychology topics	Achieved
BLOB	BLOB	Complete a psychology reflection summary	Set

Action Output:

	Time	Action	Response	Duration / Fetch Time
1	17:40:07	USE MentorLink	0 row(s) affected	0.00030 sec
2	17:40:07	SELECT * FROM Goal LIMIT 0, 1000	15 row(s) returned	0.00048 sec / 0.000...

MySQL Workbench interface showing a query execution result for the 'Rating' table.

Query executed:

```
1 USE MentorLink;
2
3 SELECT * FROM Rating;
```

Result Grid:

RatingID	MentorshipID	RaterUserID	RatedUserID	RatingValue	RatingDate
BLOB	BLOB	BLOB	BLOB	Poor	2026-03-03 16:10:00
BLOB	BLOB	BLOB	BLOB	Good	2026-03-01 10:15:00
BLOB	BLOB	BLOB	BLOB	Good	2026-03-04 12:05:00
BLOB	BLOB	BLOB	BLOB	Neutral	2026-03-05 13:50:00
BLOB	BLOB	BLOB	BLOB	Poor	2026-03-06 15:25:00
BLOB	BLOB	BLOB	BLOB	Good	2026-03-01 11:30:00
BLOB	BLOB	BLOB	BLOB	Neutral	2026-03-02 09:45:00
BLOB	BLOB	BLOB	BLOB	Good	2026-03-02 14:20:00

Sample Data Functionality

The screenshot displays a database management tool interface with a dark theme. At the top, there are tabs for 'Administration' and 'Schemas'. Below the 'Schemas' tab, a list of schemas is shown, including 'CHEMAS', 'MentorLink', 'Goal', 'Mentorship', 'MentorshipMember', 'Rating', 'Session', 'Subject', 'User', 'User_Subject', 'Views', 'Stored Procedures', 'Functions', and 'sys'. The 'Rating' schema is selected, and its table structure is shown in the 'Object Info' pane on the left. The table 'Rating' has columns: RatingID (binary(16) PK), MentorshipID (binary(16)), RaterUserID (binary(16)), RatedUserID (binary(16)), RatingValue (enum('Poor','Neutral','Good')), and RatingDate (timestamp).

The main query editor shows a SQL query:

```
USE MentorLink;  
SELECT * FROM User_Subject;
```

The query is executed, and the results are displayed in the 'Result Grid' pane. The grid shows 10 rows of data with columns: UserID, SubjectID, and UserType. The data is as follows:

UserID	SubjectID	UserType
BLOB	BLOB	Sought
BLOB	BLOB	Offered
BLOB	BLOB	Sought
BLOB	BLOB	Offered
BLOB	BLOB	Sought
BLOB	BLOB	Offered
BLOB	BLOB	Sought
BLOB	BLOB	Offered
BLOB	BLOB	Sought
BLOB	BLOB	Offered

The 'Action Output' pane at the bottom shows the execution details of the query:

	Time	Action	Response	Duration / Fetch Time
1	17:41:39	USE MentorLink	0 row(s) affected	0.00020 sec
2	17:41:39	SELECT * FROM User_Subject LIMIT 0,...	24 row(s) returned	0.00038 sec / 0.000...

Constraints

MySQL Workbench

Local instance MySQL80 - Warning - not supported | Local instance MySQL80 (mentorlink) - Warning - not supported

Administration | Schemas | subject_entity | user_entity | mentorship_entity | mentorship_member_entity | session_entity | goal_rating

SCHEMAS

Filter objects

MentorLink

- Tables
 - Goal
 - Mentorship
 - MentorshipMember
 - Rating
 - Session
 - Subject
 - User
 - User_Subject
- Views
- Stored Procedures
- Functions
- sys

Object Info | Session

Table: Rating

Columns:

- RatingID: binary(16) PK
- MentorshipID: binary(16)
- RaterUserID: binary(16)
- RatedUserID: binary(16)
- RatingValue: enum('Poor','Neutral','Good')
- RatingDate: timestamp

```
1  -- valid check
2  SELECT * FROM Rating;
3
4  -- invalid insert example
5  INSERT INTO Rating (MentorshipID, RaterUserID, RatedUserID, RatingValue, RatingDate)
6  VALUES (
7      UUID_TO_BIN('11111111-1111-1111-1111-111111111111'),
8      UUID_TO_BIN('22222222-2222-2222-2222-222222222222'),
9      UUID_TO_BIN('33333333-3333-3333-3333-333333333333'),
10     'Excellent',
11     NOW()
12 );
```

100% | 3:12

Result Grid

RatingID	MentorshipID	RaterUserID	RatedUserID	RatingValue	RatingDate
BLOB	BLOB	BLOB	BLOB	Poor	2026-03-03 16:10:00
BLOB	BLOB	BLOB	BLOB	Good	2026-03-01 10:15:00
BLOB	BLOB	BLOB	BLOB	Good	2026-03-04 12:05:00

Rating 9

Apply | Revert

Action Output

	Time	Action	Response	Duration / Fetch Time
1	17:43:21	SELECT * FROM Rating LIMIT 0, 1000	8 row(s) returned	0.00069 sec / 0.000...
2	17:43:21	INSERT INTO Rating (MentorshipID, Rat...	Error Code: 1265. Data truncated for column 'RatingV...	0.00091 sec

Example of Stored Procedure: SubmitRating

The screenshot displays the MySQL Workbench interface. The left sidebar shows the 'SCHEMAS' panel with a tree view of databases. The 'MentorLink' database is selected, and the 'Stored Procedures' folder is expanded, showing the 'SubmitRating' procedure. The main editor window shows the SQL code for creating the procedure. The code includes a comment to use the MentorLink database, a drop statement for the existing procedure, and a create statement for the new procedure. The procedure takes four parameters: p_MentorshipID, p_RaterUserID, p_RatedUserID, and p_RatingValue. It declares four variables to hold binary representations of the UUIDs and sets them using the UUID_TO_BIN function. The procedure body is currently empty, starting with a BEGIN statement.

```
1 USE MentorLink;
2
3 DROP PROCEDURE IF EXISTS SubmitRating;
4 DELIMITER $$
5
6 CREATE PROCEDURE SubmitRating (
7     IN p_MentorshipID CHAR(36),
8     IN p_RaterUserID CHAR(36),
9     IN p_RatedUserID CHAR(36),
10    IN p_RatingValue ENUM('Poor', 'Neutral', 'Good')
11 )
12 BEGIN
13     -- Convert UUID strings to BINARY(16)
14     DECLARE v_MentorshipID BINARY(16);
15     DECLARE v_RaterUserID BINARY(16);
16     DECLARE v_RatedUserID BINARY(16);
17
18     SET v_MentorshipID = UUID_TO_BIN(p_MentorshipID);
19     SET v_RaterUserID = UUID_TO_BIN(p_RaterUserID);
20     SET v_RatedUserID = UUID_TO_BIN(p_RatedUserID);
21
22
```

The bottom panel shows the 'Action Output' table, which contains the execution results of the SQL statements:

	Time	Action	Response	Duration / Fetch Time
✓	17:49:37	USE MentorLink	0 row(s) affected	0.00063 sec
▲	17:49:37	DROP PROCEDURE IF EXISTS SubmitR...	0 row(s) affected, 1 warning(s): 1305 PROCEDURE m...	0.0018 sec
✓	17:49:37	CREATE PROCEDURE SubmitRating (...	0 row(s) affected	0.0022 sec

Duplicate Rating, Self-Rating, Roles

100% 40:27

Result Grid

Filter Rows:

Export:

Result Grid

Form Editor

TestName	Result	Details
TEST 2: Wrong role (mentor -> mentee)	PASS	Only mentees can rate mentors

Result 1

Result 2

Result 3

Result 4

Result 5

>>

Read Only

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 1	17:55:25	USE MentorLink	0 row(s) affected	0.00075 sec
✓ 2	17:55:25	DROP PROCEDURE IF EXISTS SubmitR...	0 row(s) affected	0.0048 sec
✓ 3	17:55:25	CREATE PROCEDURE SubmitRating (...)	0 row(s) affected	0.0015 sec
✓ 4	17:55:36	USE MentorLink	0 row(s) affected	0.00035 sec
✓ 5	17:55:36	DROP PROCEDURE IF EXISTS RunSub...	0 row(s) affected	0.0021 sec
✓ 6	17:55:36	CREATE PROCEDURE RunSubmitRating...	0 row(s) affected	0.0023 sec

Duplicate Rating, Self-Rating, Roles

100% 40:27

Result Grid

Filter Rows:

Export:

	TestName	Result	Details
	TEST 3: Self-rating	PASS	User cannot rate themselves

Result Grid

Form Editor

Result 2

Result 3

Result 4

Result 5

Result 6

>>

Read Only

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 1	17:55:25	USE MentorLink	0 row(s) affected	0.00075 sec
✓ 2	17:55:25	DROP PROCEDURE IF EXISTS SubmitR...	0 row(s) affected	0.0048 sec
✓ 3	17:55:25	CREATE PROCEDURE SubmitRating (...)	0 row(s) affected	0.0015 sec
✓ 4	17:55:36	USE MentorLink	0 row(s) affected	0.00035 sec
✓ 5	17:55:36	DROP PROCEDURE IF EXISTS RunSub...	0 row(s) affected	0.0021 sec
✓ 6	17:55:36	CREATE PROCEDURE RunSubmitRating...	0 row(s) affected	0.0023 sec

Duplicate Rating, Self-Rating, Roles

100% 40:27

Result Grid

Filter Rows: Search

Export:

TestName	Result	Details
TEST 4: Duplicate rating	PASS	Rating already exists for this mentorship pair

Result Grid

Form Editor

Result 3

Result 4

Result 5

Result 6

Result 7

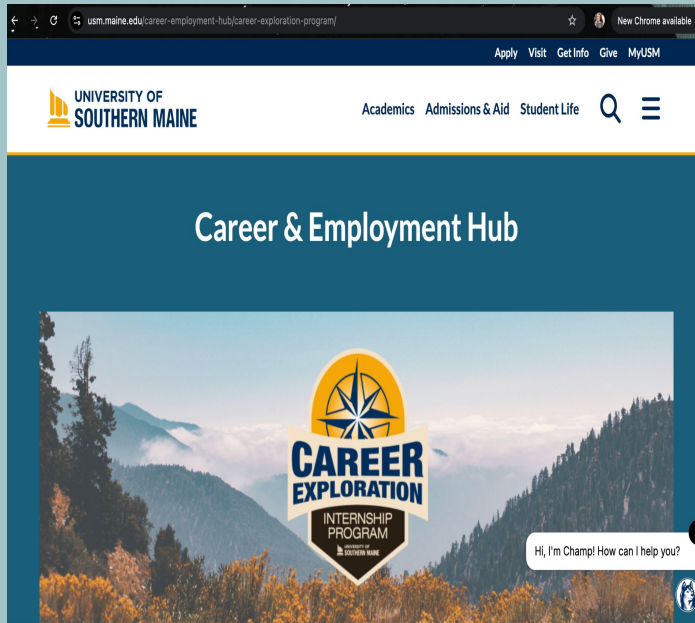
>>

Read Only

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 1	17:55:25	USE MentorLink	0 row(s) affected	0.00075 sec
✓ 2	17:55:25	DROP PROCEDURE IF EXISTS SubmitR...	0 row(s) affected	0.0048 sec
✓ 3	17:55:25	CREATE PROCEDURE SubmitRating (...)	0 row(s) affected	0.0015 sec
✓ 4	17:55:36	USE MentorLink	0 row(s) affected	0.00035 sec
✓ 5	17:55:36	DROP PROCEDURE IF EXISTS RunSub...	0 row(s) affected	0.0021 sec
✓ 6	17:55:36	CREATE PROCEDURE RunSubmitRating...	0 row(s) affected	0.0023 sec
✓ 7	17:55:36	CALL RunSubmitRatingTests()	1 row(s) returned	0.0016 sec / 0.00001

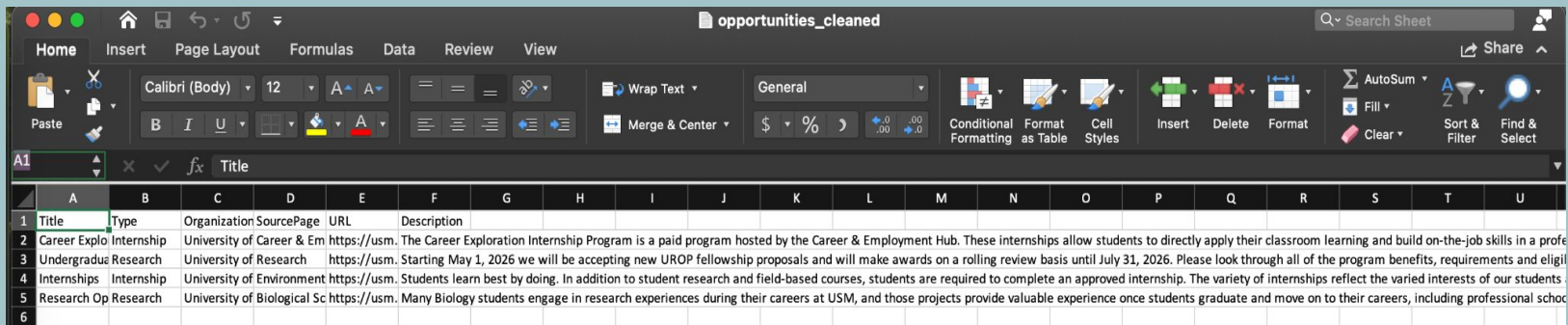
Scraping Demo



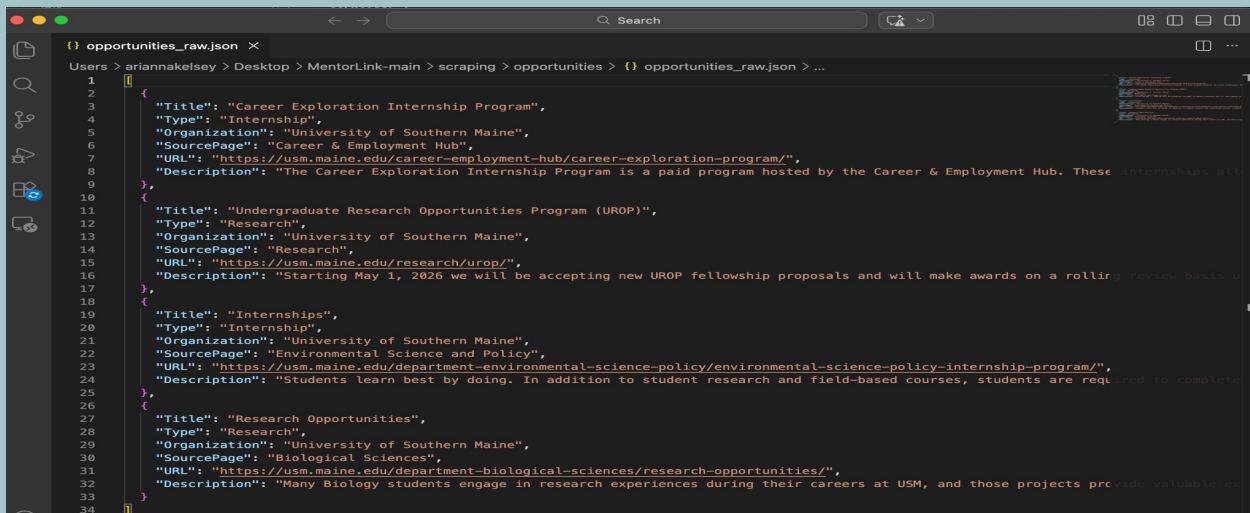
```
opportunities — -zsh — 80x24

Last login: Thu Mar 26 13:57:47 on ttys000
ariannakelsey@MacBook-Air ~ % python3 scrape_opportunities.py
/Library/Frameworks/Python.framework/Versions/3.14/Resources/Python.app/Contents
/MacOS/Python: can't open file '/Users/ariannakelsey/scrape_opportunities.py': [
Errno 2] No such file or directory
ariannakelsey@MacBook-Air ~ % cd ~/Desktop/Mentorlink-main/scraping/opportunitie
s
ariannakelsey@MacBook-Air opportunities % python3 scrape_opportunities.py
[SUCCESS] Saved 4 raw records to opportunities_raw.json
[SUCCESS] Saved 4 cleaned records to opportunities_cleaned.csv
ariannakelsey@MacBook-Air opportunities %
```

Raw v.s Cleaned Scraped Data



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	Title	Type	Organization	SourcePage	URL	Description															
2	Career Explo	Internship	University of Career & Em	https://usm.	The Career Exploration Internship Program is a paid program hosted by the Career & Employment Hub. These internships allow students to directly apply their classroom learning and build on-the-job skills in a profes																
3	Undergradua	Research	University of Research	https://usm.	Starting May 1, 2026 we will be accepting new UROP fellowship proposals and will make awards on a rolling review basis until July 31, 2026. Please look through all of the program benefits, requirements and eligi																
4	Internships	Internship	University of Environment	https://usm.	Students learn best by doing. In addition to student research and field-based courses, students are required to complete an approved internship. The variety of internships reflect the varied interests of our students																
5	Research Op	Research	University of Biological Sc	https://usm.	Many Biology students engage in research experiences during their careers at USM, and those projects provide valuable experience once students graduate and move on to their careers, including professional schoo																
6																					



```
opportunities_raw.json
Users > ariannakelsey > Desktop > MentorLink-main > scraping > opportunities > {} opportunities_raw.json > ...
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
[
  {
    "Title": "Career Exploration Internship Program",
    "Type": "Internship",
    "Organization": "University of Southern Maine",
    "SourcePage": "Career & Employment Hub",
    "URL": "https://usm.maine.edu/career-employment-hub/career-exploration-program/",
    "Description": "The Career Exploration Internship Program is a paid program hosted by the Career & Employment Hub. These internships allow students to directly apply their classroom learning and build on-the-job skills in a professional setting."
  },
  {
    "Title": "Undergraduate Research Opportunities Program (UROP)",
    "Type": "Research",
    "Organization": "University of Southern Maine",
    "SourcePage": "Research",
    "URL": "https://usm.maine.edu/research/urop/",
    "Description": "Starting May 1, 2026 we will be accepting new UROP fellowship proposals and will make awards on a rolling review basis until July 31, 2026. Please look through all of the program benefits, requirements and eligibility criteria."
  },
  {
    "Title": "Internships",
    "Type": "Internship",
    "Organization": "University of Southern Maine",
    "SourcePage": "Environmental Science and Policy",
    "URL": "https://usm.maine.edu/departments-environmental-science-policy/environmental-science-policy-internship-program/",
    "Description": "Students learn best by doing. In addition to student research and field-based courses, students are required to complete an approved internship. The variety of internships reflect the varied interests of our students."
  },
  {
    "Title": "Research Opportunities",
    "Type": "Research",
    "Organization": "University of Southern Maine",
    "SourcePage": "Biological Sciences",
    "URL": "https://usm.maine.edu/departments-biological-sciences/research-opportunities/",
    "Description": "Many Biology students engage in research experiences during their careers at USM, and those projects provide valuable experience once students graduate and move on to their careers, including professional school."
  }
]
```

Indexing/Query Optimization

The screenshot shows a database IDE interface. At the top, there are tabs for 'session_entity', 'goal_rating', 'SQL File 19*', 'opportunities_query', and 'SQL File 28*'. Below the tabs is a toolbar with various icons and a 'Limit to 1000 rows' dropdown. The main editor area contains SQL code:

```
42
43
44 -----
45 -- CREATE INDEX
46 -- -----
47 CREATE INDEX idx_opportunity_type
48 ON Opportunity(Type);
49
50 -----
```

Below the editor, there is a 'Result Grid' section with a search bar and an 'Export' button. The grid shows the following result:

EXPLAIN
-> Index lookup on Opportunity using idx_opport...

Below the result grid, there are tabs for 'Result 5', 'Result 6', 'Result 7', and 'Opportunity 8'. The 'Opportunity 8' tab is selected, and it shows a 'Read Only' status.

Below the tabs, there is an 'Action Output' section with a table showing the execution log:

	Time	Action	Response	Duration / Fetch Time
✓ 1	18:09:28	USE MentorLink	0 row(s) affected	0.00066 sec
✓ 2	18:09:28	DROP TABLE IF EXISTS Opportunity	0 row(s) affected	0.012 sec
✓ 3	18:09:28	CREATE TABLE Opportunity (Opportunity...	0 row(s) affected	0.0088 sec
✓ 4	18:09:28	INSERT INTO Opportunity (Title, Type,...	10 row(s) affected Records: 10 Duplicates: 0 Warnin...	0.0017 sec
✓ 5	18:09:28	EXPLAIN ANALYZE SELECT Opportunit...	1 row(s) returned	0.00038 sec / 0.000...
✓ 6	18:09:28	CREATE INDEX idx_opportunity_type O...	0 row(s) affected Records: 0 Duplicates: 0 Warnings...	0.016 sec
✓ 7	18:09:28	EXPLAIN ANALYZE SELECT Opportunit...	1 row(s) returned	0.00034 sec / 0.000...
✓ 8	18:09:28	SELECT Type, COUNT(*) AS Oppo...	3 row(s) returned	0.00029 sec / 0.0000...
✓ 9	18:09:28	SELECT Title, Type, Organizatio...	9 row(s) returned	0.00038 sec / 0.000...